K23843660: BIG-IP LTM profiles | BIG-IP LTM-DNS operations guide

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Chapter 5: BIG-IP LTM profiles

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Profiles allow the BIG-IP LTM system to understand or interpret supported network traffic types and affect the behavior of managed traffic flowing through it.

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Standard profiles

BIG-IP systems include a set of standard (default) profiles for a wide array of applications and use cases, such as performing protocol enforcement of traffic, enabling connection and session persistence, and implementing client authentication.

The following table lists common default BIG-IP LTM profiles:

Table 5.1 Common BIG-IP LTM profiles

Profile type	Details
Service	L7 protocols, such as HTTP, FTP, DNS, and SMTP.
Content	Content HTML profile
Persistence	Settings govern routing of existing or related connections or requests.
Protocol	L4 protocols, such as TCP, UDP, and Fast L4.
SSL	Enable the interception, offloading, and authentication of transport layer encryption.
Message Routing	Configure message routing protocols, such as Diameter, SIP, and MQTT.
Other	Provide various types of advanced functionality.

For more information about BIG-IP protocol profiles, refer to **Protocol Profiles** in ***BIG-IP Local Traffic Management: Profiles Reference***.

***Note**: For information about how to locate F5 product manuals, refer to **K98133564: Tips for searching AskF5 and finding product documentation**.*

Profile management

F5 recommends the following guidelines for working with default profiles:

- Changes to the default profile settings are lost when the BIG-IP system software is upgraded. Make sure that data is backed up before upgrading.
- Create a child profile of the default profile and make custom updates to the child rather than altering the default profile itself. Changes to default profiles may have unexpected consequences for custom profiles that inherit default settings.

***Note**: A custom profile can inherit settings from another custom profile. This allows for efficient management of profiles for related applications requiring similar-but-unique configuration options.*

Protocol profiles

There are a number of profiles available that are jointly described as protocol profiles. For more information on the following profiles, refer to **Protocol Profiles** in ***BIG-IP Local Traffic Manager: Profiles Reference***, along with the referenced solutions.

***Note**: For information about how to locate F5 product manuals, refer to **K98133564: Tips for searching AskF5 and finding product documentation**.*

Fast L4

FastL4 profiles are used for Performance (Layer 4), Forwarding (Layer 2), and Forwarding (IP) virtual servers. The FastL4 profile can also be used to enable stateless IP level forwarding of network traffic in the same fashion as an IP router. For more information, refer to AskF5 article: **K7595: Overview of IP forwarding virtual servers**.

Fast HTTP

The FastHTTP profile is a scaled-down version of the HTTP profile optimized for speed under controlled traffic conditions. It can only be used with the Performance HTTP virtual server and is designed to speed up certain types of HTTP connections and reduce the number of connections to servers.

Because the FastHTTP profile is optimized for performance under ideal traffic conditions, the HTTP profile is recommended when load balancing most general-purpose web applications.

Refer to **K8024: Overview of the FastHTTP profile** before deploying performance (HTTP) virtual servers.

TCP

TCP profiles are configuration tools that help you to manage TCP network traffic. Many of the configuration settings of TCP profiles are standard SYSCTL types of settings, while others are unique to the BIG-IP system.

TCP profiles are important because they are required for implementing certain types of other profiles. For example, by implementing TCP, HTTP, Rewrite, HTML, and OneConnect profiles, along with a persistence profile, you can take advantage of various traffic management features, such as:

- Content spooling, to reduce server load
- OneConnect, to pool idle server-side connections
- Layer 7 session persistence, such as hash or cookie persistence

- iRules for managing HTTP traffic
- HTTP data compression
- HTTP pipelining
- URI translation
- HTML content modification
- Rewriting of HTTP redirections

The BIG-IP system includes several pre-configured TCP profiles that you can use as is. In addition to the default tcp profile, the system includes TCP profiles that are pre-configured to optimize LAN and WAN traffic, as well as traffic for mobile users. You can use the pre-configured profiles as is, or you can create a custom profile based on a pre-configured profile and then adjust the values of the settings in the profiles to best suit your particular network environment.

TCP base profiles

In BIG-IP 13.0 and later, several new TCP base profiles are available in addition to the existing profiles, which have been updated and are now read-only.

Table 5.2 BIG-IP 13.0 and later base profiles

TCP profile	Description
tcp	Existing. Provides “safe” new features. Serves as parent to all new F5 base profiles.
tcp-legacy	Existing. Derived from tcp profile. Serves as parent to all previously existing base profiles.
f5-tcp-progressive	New. Most performance features enabled.
f5-tcp-lan	New. Loosely based on tcp-lan-optimized, which is deprecated in BIG-IP 13.0 and later.
f5-tcp-wan	New. Loosely based on tcp-wan-optimized, which is deprecated in BIG-IP 13.0 and later.
f5-tcp-mobile	New. Loosely based on tcp-mobile-optimized, which is deprecated in BIG-IP 13.0 and later.

TCP autotune

In BIG-IP 13.0 and later, the TCP profile includes an autotune feature which allows you to autotune TCP send and receive buffers as well as the proxy buffer used to interface with the rest of the connection flow.

For general information on the TCP profile, refer to **K50411377: Overview of the TCP profile (16.x)**.

UDP

Use of a user datagram protocol (UDP) profile is mandatory on standard UDP virtual servers.

For more detail on the UDP profile refer to **K7535: Overview of the UDP profile**.

SCTP

Stream control transmission protocol (SCTP) is a L4 transport protocol, designed for message-oriented applications that transport signaling data, such as Diameter.

OneConnect

The OneConnect profile may improve HTTP performance by reducing connection setup latency between the BIG-IP system and pool members, as well as minimizing the number of open connections to them.

The OneConnect profile maintains a pool of connections to the configured pool members. When there are idle connections available in the connection pool, new client connections uses these existing pool member connections if the configuration permits. When used in conjunction with an HTTP profile, each client request from an HTTP connection is load-balanced independently.

For information about settings within the OneConnect profile, refer to **About OneConnect Profiles** in ***BIG-IP Local Traffic Management: Profiles Reference***.

Note*: For information about how to locate F5 product manuals, refer to* ***K98133564: Tips for searching AskF5 and finding product documentation****.*

OneConnect use cases

HTTP requests can be load-balanced to different pool members on a request-by-request basis by enabling OneConnect on a virtual server. When using an HTTP profile without a OneConnect profile, the BIG-IP system makes a load balancing decision based on the first request received on the connection, and all subsequent requests on that connection continues to go to the same pool member.

iRules can be used to select pools and pool members based on information contained in each request, such as the URI path or user-agent string.

The OneConnect source mask setting manages connection re-use, and is applied to the client address on the server-side of a connection to determine its eligibility for connection re-use. In this way, it is possible to limit which clients can share server-side connections with each other.

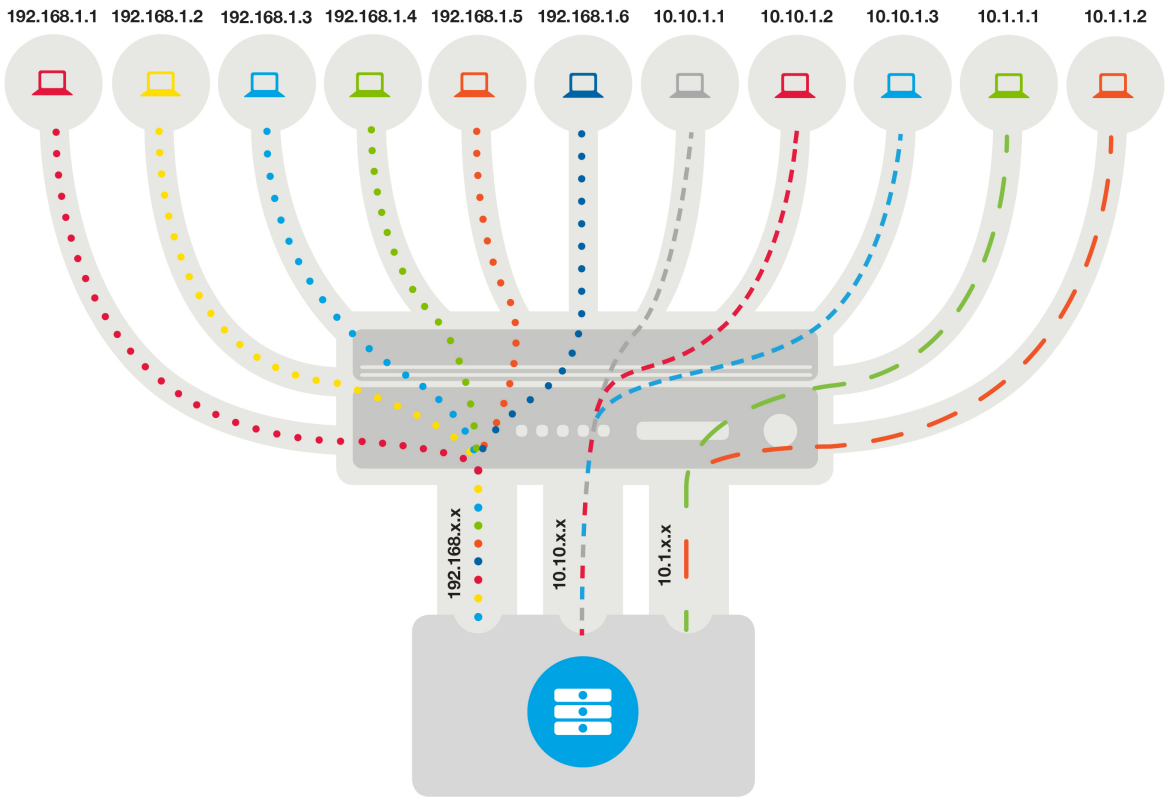

Figure 5.1 OneConnect source mask connection reuse

Refer to **K85357500: Managing connection reuse using OneConnect source mask (12.x - 16.x)**.

OneConnect limitations

F5 recommends careful consideration of the following limitations and implications before using a OneConnect profile:

- OneConnect is intended to work with traffic that can be inspected by the BIG-IP system at L7.
- OneConnect requires the use of an HTTP profile to process HTTP traffic.
- When using the default OneConnect profile, the pool member cannot rely on the client IP address information in the network headers to accurately represent the source of the request. Consider using the X-Forwarded-For option in an HTTP profile to pass the client IP address to the pool members using an HTTP header.
- Applications that use connection-based authentication such as NT Lan Manager (NTLM), may need additional profiles or may not be compatible with OneConnect. For more information, refer to **K10477: Optimize NTLM traffic in BIG-IP 10.x or later** or **Other Profiles** in ***BIG-IP Local Traffic Management: Profiles Reference***.

Note: *For information about how to locate F5 product manuals, refer to* ***K98133564: Tips for searching AskF5 and finding product documentation***.

- Before using OneConnect with non-HTTP applications, refer to **K7208: Overview of the OneConnect profile**.

HTTP profiles

BIG-IP LTM includes several profile types used for optimizing or augmenting HTTP traffic, including the following:

- **HTTP**
- **HTTP/2**
- **QUIC and HTTP/3 Gateway**
- **HTTP compression**
- **Web acceleration**

HTTP

The HTTP profile enables the use of HTTP features in BIG-IP LTM policies and iRules. It is required for using other HTTP profile types, such as HTTP compression and web acceleration.

In BIG-IP 11.5 and later, HTTP profiles can operate in one of the following proxy modes:

Table 5.3 HTTP proxy modes

Proxy	Details
Reverse	You can specify the Reverse Proxy Mode to enable the BIG-IP system to manage responses from multiple servers.
Explicit	The Explicit Proxy Mode enables the BIG-IP system to handle HTTP proxy requests and function as a gateway. By configuring browser traffic to use the proxy, you can control whether to allow or deny a requested connection, based on configured policies. The Explicit Proxy Mode requires a DNS resolver, specified in the Explicit Proxy area of the screen.
Transparent	The Transparent Proxy Mode enables the BIG-IP system to forward invalid HTTP traffic to a specified server, instead of dropping the connection. By configuring an HTTP profile to forward invalid HTTP traffic, you can manage various atypical service provider scenarios, such as HTTP traffic from non-browser clients that function as web browsers.

Here are some of the most common options:

- Response chunking changes the behavior of BIG-IP LTM handling of a chunked response from the server. By default it is set to Selective, which reassembles and then re-chunk chunked responses while preserving

unchunked responses. This is the recommended value. However, setting this to Unchunk may be required if clients are unable to process chunked responses.
- Connect Transformations controls whether BIG-IP LTM modifies Connection: Close headers sent in response to HTTP/1.0 requests to Keep-Alive. This is enabled by default. It allows HTTP/1.0 clients to take advantage of OneConnect.
- Redirect Rewrite allows BIG-IP LTM to modify or strip redirects sent by pool members in order to prevent clients from being redirected to invalid or unmanaged URLs. This defaults to None, which disables the feature.
- Insert X-Forwarded-For causes BIG-IP LTM to insert an X-Forwarded-For HTTP header in requests. This reveals the client IP address to servers even when source address translation is used. By default, this is Disabled.
- HTTP Strict Transport Security (HSTS) causes BIG-IP LTM 12.0 and later to insert the RFC6797 STS header into the server response back to the client. For more information about HSTS, refer to the **OWASP website** (https://www.owasp.org). Disabled by default.

***Note**: This link take you to resources outside of AskF5, and it is possible that the documents may be removed without our knowledge.*

For more information, refer to **HTTP Profiles** in ***BIG LTM Concepts***.

***Note**: For information about how to locate F5 product manuals, refer to **K98133564: Tips for searching AskF5 and finding product documentation**.*

HTTP/2

The HTTP/2 profile (BIG-IP 11.6.0 and later) allows the BIG-IP system to serve as a gateway for HTTP/2 traffic. By multiplexing streams and compressing headers, the perceived latency of requests and responses is reduced and the overall efficiency of the network is improved. You can use the HTTP/2 profile can to provide the following:

- Multiplexed request/response streams with flow control for improved network usage.
- Automatic header compression.
- Binary instead of textual message framing for efficient message processing.
- Support for SPDY, HTTP/1.1, and HTTP/2 protocol selection.
- Proactive server response push to client.
- iRules logic for HTTP/2.

Limitations of the HTTP/2 profile

- Header compression consumes CPU and memory resources.
- No support for source address persistence prior to BIG-IP 15.0.0.
- Not compatible with NT LAN Manager (NTLM) protocols.
- Not compatible with SSL profile (Client) renegotiation.

QUIC and HTTP/3 Gateway

In BIG-IP 15.1.0 and later, you can proxy application servers built on HTTP or HTTP/2 to communicate with clients using HTTP/3 over QUIC. HTTP/3 offers similar APIs to HTTP/2 while taking advantage of the new QUIC transport protocol, which replaces most of the functions of TCP while offering the following advantages:

- **Lower latency**: Broader and more robust support for "0-RTT" application data, allowing that data exchange to occur in the first round trip.
- **Better loss response**: QUIC has a number of improvements to allow data transfer to occur more quickly in the presence of packet loss.
- **Stronger privacy properties**: QUIC exposes much less information about the connection to untrusted middle boxes.

- **Better extensibility**: The TCP architecture contains several impediments to true innovation in the transport layer that may greatly improve user experience. QUIC protocol's privacy and version numbering capabilities make it much easier to deploy an evolving protocol optimized for the internet of tomorrow.
- **Address migration**: Although the BIG-IP system does not yet implement this feature, QUIC has robust support for clients that change their IP address in mid-connection.

The QUIC and HTTP/3 specifications are still draft standards, but F5 expects them to become full standards later in 2020. BIG-IP 15.1.0.1 interoperates with draft-24 of the standard, and future maintenance releases will keep the draft compliance up-to-date.

HTTP Compression profile

An HTTP Compression profile allows BIG-IP LTM to compress HTTP response content, which can substantially reduce network traffic. Although data compression can also be performed by most web servers, using BIG-IP LTM to compress traffic for load balanced applications can help minimize server load and increase total capacity.

If the web server has already compressed the data, the BIG-IP system cannot perform traffic inspection.

***Note**: F5 recommends reviewing AskF5 and/or manuals regarding the options available for HTTP compression profiles before implementing this feature. However, the default profile compresses most text including HTML, JavaScript, and XML data. Most image data cannot be further compressed.*

The following table lists some common HTTP Compression profile options and their details.

Table 5.4 Common HTTP Compression profile options

Option	Details
Keep Accept Encoding	Disabled by default. Enable only if you want server to compress data instead of having BIG-IP LTM do it. Default behavior is to strip the Accept-Encoding header from requests, which prevents HTTP servers from compressing responses.
HTTP/1.0 Requests	Disabled by default. Allows BIG-IP LTM to compress responses to HTTP/1.0 requests.
CPU Saver	Enabled by default. (Recommended.) Allows BIG-IP LTM to reduce compression levels and even disable compression when CPU use reaches a predefined threshold.
gzip	Allows you to adjust the compression level. Higher values increase compression efficiency in exchange for a higher latency and higher resource cost in both CPU and memory. Default settings balance resource use, allow minimal latency, and promote effective compression.

For more information about HTTP compression profile options, refer to **K15434: Overview of the HTTP Compression profile**.

Web Acceleration profile

A Web Acceleration profile enables caching behavior in BIG-IP LTM, according to guidelines set in RFC 2616. Web acceleration is highly recommended if you are using HTTP compression and have a large amount of static content, as BIG-IP LTM can cache the compressed response, resulting in reduced BIG-IP LTM and server load.

F5 recommends customizing cache settings to maximize retention of static data. For more information, refer to **K14903: Overview of the Web Acceleration profile**.

SSL Profiles

BIG-IP LTM supports encrypting both the client-side and server-side flows on full proxy virtual servers. Client SSL profiles encrypt and decrypt the client-side flow (BIG-IP LTM acts as the SSL server), and server SSL profiles encrypt and decrypt the server-side flow (BIG-IP LTM acts as the SSL client).

When no SSL profile is enabled on a virtual server but the pool members expect SSL, it is called SSL Passthrough and no L7 profiles may be used. A FastL4 virtual server may be preferred for maximum performance in this instance.

SSL Offload

Figure 5.2 SSL offload

In SSL offload mode, only a client SSL profile is used. While the connection between the client and the BIG-IP system is encrypted, the connection from BIG-IP LTM to the pool members is unencrypted. This completely removes the requirement for the pool members to perform SSL encryption and decryption, which can reduce resource usage on pool members and improve overall performance.

SSL offload should only be used on trusted, controlled networks.

***Note**: The BIG-IP Remote Crypto Offload feature may be limited in performance due to multiple factors, which can include licensing and remote host hardware capabilities.*

SSL re-encryption

Figure 5.3 SSL re-encryption

In SSL re-encryption mode, both client SSL and server SSL profiles are configured. The connection to pool members is also encrypted. This requires that the pool members perform SSL encryption and decryption as well, but offers security on the server-side flow.

Server SSL profiles can also be configured with a client certificate to authenticate using SSL client certificate auth to the pool member. This method can be used to ensure that the service can only be accessed using BIG-IP LTM virtual server, even if a client can initiate connections directly to the pool members.

You can use an server SSL profile without a client SSL profile, but the client-side connection is not encrypted.

Other SSL features

- Proxy SSL allows clients to do client certificate authentication directly to the pool member while still allowing BIG-IP LTM to inspect the traffic.
- SSL Forward Proxy (BIG-IP 11.5.0 and later) allows BIG-IP LTM to intercept SSL traffic destined for external third-party servers. For more information refer to the relevant sections of ***BIG-IP Local Traffic Manager: Concepts*** and ***BIG-IP Local Traffic Manager: Monitors Reference*** for your version.

Note*: For information about how to locate F5 product manuals, refer to* ***K98133564: Tips for searching AskF5 and finding product documentation****.*

SSL profile options

The following table lists some common SSL profile options and their details.

Table 5.5 Common SSL profile options

Option	Details
Certificate	Specifies the certificate BIG-IP LTM offers. On client SSL profiles, this must be a server certificate; on server SSL, it must be a client certificate.
Key	Specifies the key which is used for authentication purposes. It must match the configured certificate. Instructions to verify this are available on AskF5. When attempting to configure a certificate and key, BIG-IP LTM generates an error if they do not match.
Chain (ClientSSL only)	Allows you to configure an intermediate or chain certificate that BIG-IP LTM presents in addition to its own certificate. Contact your certificate authority to determine whether you need this.
Passphrase	Allows you to specify the passphrase if you are using an encrypted key. This increases security because if the key is lost it is still not compromised. The passphrase is also encrypted in the configuration using the SecureVault TMOS feature.
Ciphers	Specifies which ciphers are enabled for use by this SSL profile in an OpenSSL-format cipher list. Defaults vary between client and server and between versions.
Options	May be used to selectively disable different SSL and TLS versions, if required by your security policy.
Client Authentication (client SSL only)	Contains parameters which allow BIG-IP LTM to verify certificates presented by an SSL client.
Server Authentication (server SSL only)	Contains parameters which allow BIG-IP LTM to validate the server certificate; normally, it is ignored.

Note*: On client SSL profiles in BIG-IP 11.6 and later, the Certificate Key Chain section must be saved by selecting* ***Add****. You can now configure multiple key and certificate pairs (including a chain certificate, if required) per SSL profile: one for RSA and one for DSA.*

For a complete list of SSL profiles and their options, refer to : **K14783: Overview of the Client SSL profile (11.x - 17.x)** and **K14806: Overview of the Server SSL profile (11.x - 17.x)**.

Maximum Aggregate Renegotiation option

In BIG-IP 12.0 and later, you can use a0 **Maximum Aggregate** setting for renegotiation to mitigate DDoS attacks.

When a client connects to an SSL server, a DDoS attack can occur by the client sending requests for renegotiation of parameters. Because the server has to work harder than the client, it is possible for a malicious attack to over-burden the server to the point that its resources can no longer accept any more traffic, good or bad.

In earlier versions F5 provides the ability to protect against such attacks by limiting the number of renegotiations per connection; however, an attacker could use multiple connections to generate the DDoS attack, staying under the max for the setting in each of those connections.

Recommendation is to estimate a conservative baseline for maximum number of what the box can tolerate under its workload and then set the aggregate number to leave some ceiling available. This feature drops all SSL connections that are renegotiating above that limit, good or bad. A legitimate client issuing a single renegotiation request are dropped if malicious clients have already pushed the renegotiation count above the limit. This feature is a last resort to preserve the system rather than punishing the attackers.

In BIG-IP 12.0 and later, with the **Maximum Aggregate Renegotiation** option, you can set a cumulative number of parameter renegotiations possible from all profiles. The option is disabled by default. F5 recommends talking with an F5 Consultant Service before enabling it.

Cipher strings and SSL/TLS version

For information on specifying SSL/TLS cipher strings for use with SSL profiles, refer to following :

- **K86554600: SSL ciphers supported on BIG-IP platforms (15.x)**
- **K15194: Overview of BIG-IP SSL/TLS cipher suite**
- **K8802: Using SSL ciphers with BIG-IP Client SSL and Server SSL profiles**

BIG-IP LTM policies

In BIG-IP 11.4.0 and later, BIG-IP LTM policies supersede HTTP class profiles. BIG-IP LTM policies allow functionality previously only available through iRules.

BIG-IP LTM policies are assigned to virtual servers on the Resources page under General Properties and Rules.

General properties

The most important settings in **General Properties** are **Requires** and **Controls**.

- **Requires** allows you to specify what information the policy is able to use to make decisions.
- **Controls** allows you to specify the types of actions it is able to take.

Some selections from **Requires** and **Controls** require you to assign particular profiles to virtual servers which use this policy. For example, a **Requires** setting of **http** depends on an HTTP profile, and a **Controls** setting of **server-ssl** depends on a server SSL profile.

***Note**: In BIG-IP 12.0.1 and later, **Requires** and **Controls** are automatically configured.*

Rules

Each Rule is composed of **Conditions** and **Actions**.

- **Conditions** define the requirements for the rule to run.
- **Actions** define the action taken if the **Conditions** match.

By default, the first Rule with matching **Conditions** is applied. This behavior can be changed using the **Strategy** setting under **General Properties**. For in-depth information about this behavior, refer to **K04597703: Overview of the Local Traffic Policies feature (12.1.0 and later)**.

A common use case for BIG-IP LTM policies is to select a pool based on HTTP URI. For this you need a BIG-IP LTM policy that Requires http and **Controls forwarding**. You can then configure Rules with an operand of **http-uri** and an **Action** with a target of forward to a pool.

BIG-IP LTM policies offer room for expansion: Rules can have multiple operands (all must match for the rule to match) or actions. Using the previous example, if some pools for a virtual server require ServerSSL while others do not, you could add **Controls** and Action of **server-ssl** to disable ServerSSL when it is not needed. You can also introduce advanced HTTP caching or compression behavior using BIG-IP LTM policies. For example, if some pools are just for static content, you may want to enable caching only for those pools.

BIG-IP LTM policies offer a broad range of capabilities, and new versions frequently introduce additional capabilities. For more information, refer to ***BIG-IP LTM Concepts***.

Note*: For information about how to locate F5 product manuals, refer to* ***K98133564: Tips for searching AskF5 and finding product documentation****.*

Persistence profiles

Many applications served by BIG-IP LTM are session-based and require the client to be load balanced to the same pool member for the duration of that session. BIG-IP LTM can accomplish this requirement through persistence.

When a client connects to a virtual server for the first time, a load balancing decision is made, then the configured persistence record is created for that client. All subsequent connections that the client makes to that virtual server are sent to the same pool member for the life of that persistence record.

Common persistence types

Cookie persistence uses an HTTP cookie stored on a client's computer to allow the client to reconnect to the same server previously visited at a web site. Because a cookie is an object of the HTTP protocol, use of a cookie persistence profile requires that the virtual server also have an HTTP profile assigned. The cookie persistence profile works in the following methods:

Cookie insert

BIG-IP LTM inserts an additional cookie into the response of a pool member. By default, the cookie has no explicit expiration time, making it valid for the life of the browser session. This is the most common and seamless method of cookie persistence to deploy. For more information on the cookie insert method refer to **K6917: Overview of BIG-IP persistence cookie encoding**.

Note*: Unencrypted cookies may be subject to unintended information disclosure (this is the default behavior). If the security polices of your organization require this cookie value to be encrypted, refer to* ***K14784: Configuring BIG-IP cookie encryption (10.x - 15.x)****.*

Cookie passive and cookie rewrite

With these methods, the pool member must provide all or part of the cookie contents. Properly implementing the cookie passive and cookie rewrite method requires close collaboration between the BIG-IP LTM administrator and the application owner.

Source address affinity

Directs session requests to the same pool member based on the source IP address of a packet. This profile can also be configured with a network mask so that multiple clients can be grouped into a single persistence method. The network mask can also be used to reduce the amount memory used for storing persistence records. The defaults for this method are a timeout of 180 seconds and a mask of 255.255.255.255 (one persistence record per source IP). For more information, refer to **K5779: BIG-IP LTM memory allocation for Source Address Affinity persistence records**.

Universal persistence directs session requests to the same pool member based on customizable logic written in an iRules. This persistence method is commonly used in many BIG-IP iApps.

Other persistence types

The following persistence methods are also available.

SSL persistence

SSL persistence tracks non-terminated SSL sessions using the SSL session ID. Even when the client's IP address changes, BIG-IP LTM still recognizes the connection as being persistent based on the session ID. This method can be problematic when working with clients who frequently renegotiate the SSL session.

For more information, refer to **K3062: Using SSL session ID persistence**.

Destination address affinity

Destination address affinity directs session requests to the same pool member based on the destination IP address of a packet. This method is commonly used for load balancing outbound traffic over dual WAN circuits or to multiple network devices. This method should not be used when load balancing application traffic to load balance pools.

Hash persistence

Hash persistence is similar to universal persistence, except that the persistence key is a hash of the data, rather than the data itself. Hash persistence may be useful when it is necessary to obfuscate the data that is being persisted upon.

Microsoft Remote Desktop Protocol persistence

Microsoft Remote Desktop Protocol (MSRDP) persistence tracks sessions between clients and pool members based on either a token provided by a Microsoft Terminal Services Session Directory/TS Session Broker server or the username from the client.

***Note**: Since the routing tokens may end up identical for some clients, the BIG-IP system may persist the RDP sessions to the same RDP servers.*

You can use Session Initiation Protocol (SIP) persistence for servers receiving SIP messages sent through UDP, SCTP, or TCP.

Persistence performance considerations

Each persistence method has a different performance implication. Some methods require the BIG-IP system to maintain an internal state table, which consumes memory and CPU resources. Other persistence methods may not be granular enough to ensure uniform distribution among pool members. For example, the source address affinity persistence method may make it difficult to uniquely identify clients on corporate networks whose source addresses are masked by firewalls and proxy servers.

CARP hash persistence

The source address, destination address, and hash persistence profiles support a Cache Array Routing Protocol (CARP) hash algorithm. When using CARP hash, the BIG-IP system performs a computation with the persistence key and each

pool member to deterministically choose a pool member. Given a set of available pool members, a client connection is always directed to the same pool member. If a pool member becomes unavailable, the sessions to that pool member is distributed among the remaining available pool members, but sessions to the available pool members remains unaffected.

CARP advantages

- The BIG-IP system maintains no state about the persistence entries, so CARP does not increase memory utilization.
- There is no need to mirror persistence. Given a persistence key and the same set of available pool members, two or more BIG-IP systems reaches the same conclusion.
- Persistence does not expire: Given the same set of available pool members, a client is always directed to the same pool member.

CARP disadvantages

- When a pool member becomes available (either due to addition of a new pool member or change in monitor state), new connections from some clients are directed to the newly available pool member at a disproportionate rate.

For more information about CARP hash, refer to **K11362: Overview of the CARP hash algorithm**.

For more information on the configuration for persistence profiles, refer to ***BIG-IP Local Traffic Manager: Concepts*** for your BIG-IP version.

Note: *For information about how to locate F5 product manuals, refer to* ***K98133564: Tips for searching AskF5 and finding product documentation***.

Other protocols and profiles

To manage application layer traffic, you can use any of the following profile types. For more information refer to ***BIG-IP Local Traffic Manager: Profiles Reference***.

Table 5.6 Profile types

Profile type	Details
File Transfer Protocol (FTP) profile	Allows modifying a few FTP properties and settings to your specific needs.
Domain Name System (DNS) profile	Allows users to configure various DNS attributes and allow for many the BIG-IP system DNS features such as DNS caching, DNS IPv6 to IPv4 translation, DNSSEC, etc.
Real Time Streaming Protocol (RTSP)	A network control protocol designed for use in entertainment and communications systems to control streaming media servers.
Internet Content Adaptation Protocol (ICAP)	Used to extend transparent proxy servers, thereby freeing up resources and standardizing the way in which new features are implemented.
Request Adapt or Response Adapt profile	Instructs an HTTP virtual server to send a request or response to a named virtual server of type Internal for possible modification by an Internet Content Adaptation Protocol (ICAP) server.
	An enhanced version of the Remote Authentication Dial-In User Service (RADIUS) protocol.

Diameter	When you configure a Diameter profile, the BIG-IP system can send client-initiated Diameter messages to load balancing servers.
Remote Authentication Dial-In User Service (RADIUS)	Used to load balance RADIUS traffic.
Session Initiation Protocol (SIP)	A signaling communications protocol, widely used for controlling multimedia communication sessions, such as voice and video calls over Internet Protocol (IP) networks.
Simple Mail Transfer Protocol (SMTP)	Secures SMTP traffic coming into the BIG-IP system. When you create an SMTP profile, BIG-IP Protocol Security Manager provides several security checks for requests sent to a protected SMTP server.
SMTPS profile	Provides a way to add SSL encryption to SMTP traffic quickly and easily.
iSession profile	Tells the system how to optimize traffic. Symmetric optimization requires an iSession profile at both ends of the iSession connection.
Rewrite profile	Instructs BIG-IP LTM to display websites differently on the external network than on an internal network and can also be used to instruct the BIG-IP system to act as a reverse proxy server.
Extensible Markup Language (XML)	Causes the BIG-IP system to perform XML content-based routing requests to an appropriate pool, pool member, or virtual server based on specific content in an XML document.
Speedy (SPDY)	An open-source web application layer protocol developed by Google in 2009 and is primarily geared toward reducing Web page latency. By using the SPDY profile, the BIG-IP system can enhance functionality for SPDY requests.
Financial Information Exchange (FIX)	An electronic communications protocol for international real-time exchange of information related to the securities transactions and markets.
Video Quality of Experience (QOE) profile	Allows assessment of an audience's video session or overall video experience, providing an indication of application performance.

Troubleshooting

To resolve most traffic-related issues, review the virtual server and networking-level configurations. These may include pools and address translation settings. You may also need to review the following:

- Architectural design for the virtual server in question to verify the correct profile is used.
- Custom profiles and their associated options to ensure they modify traffic behavior as anticipated.
    - Next, revert to an unmodified default profile and observe changes in behavior.

BIG-IP LTM-DNS operations guide

- **Chapter 1: Guide introduction and contents**
- **Chapter 2: BIG-IP LTM Load Balancing**
- **Chapter 3: BIG-IP LTM network address objects**
- **Chapter 4: BIG-IP LTM Virtual Servers**
- **Chapter 6: BIG-IP DNS/DNS Services**
- **Chapter 7: iRules**
- **Chapter 8: Logging**

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